

AWS Solutions Architect Associate

Study Notes: Cloud Fundamentals & IAM (Identity and Access Management)

PART 1 · CLOUD FUNDAMENTALS

What Is AWS?

A cloud provider that offers servers and services that can be used **on demand** and scaled easily — no need to own or manage physical hardware.

How to Choose an AWS Region

- **Compliance:** Governance and legal requirements may restrict where data can live.
- **Proximity to customers:** Pick a region close to users to reduce latency.
- **Available services:** Not every service is offered in every region — choose one that has what you need.
- **Pricing:** Costs vary from region to region.

Availability Zones (AZs)

- Each region has a **minimum of 3** and a **maximum of 6** AZs.
- Each AZ is one or more data centers with redundant power, networking, and connectivity.
- AZs are separate from each other — isolated from disasters affecting another AZ.
- AZs within a region are connected with high-bandwidth, ultra-low-latency networking.

PART 2 · IAM CORE CONCEPTS

IAM Overview

- IAM is a **global service** — it is not tied to a single region.
- **Users** represent people, and can be organized into groups.
- **Groups** can only contain users — a group cannot contain another group.
- A user doesn't have to belong to just one group — they can belong to multiple groups.
- Users and groups can be assigned permissions via JSON documents called **policies**.

■ Key Exam Concept: Least Privilege

Always grant only the permissions required to perform a task — nothing more. This principle underlies almost every IAM best-practice question on the exam.

Inline vs. Inherited Policies

Policy Type	Description
Inline Policy	Attached directly to a single, specific user only.
Inherited Policy	Attached to a group, and therefore applies to (is inherited by) every user in that group.

IAM Policy Structure

A policy document is made up of the following top-level elements:

Element	Description
Version	The policy language version.
Id	Optional identifier for the policy.
Statement	One or more individual permission statements (see below).

Each **Statement** within a policy consists of:

Field	Description
Sid	Optional statement ID.
Effect	Whether the statement allows or denies access — Allow / Deny.
Principal	Who the policy applies to: an account, user, or role.
Action	A list of actions this statement allows or denies.
Resource	A list of resources the actions apply to.
Condition	Optional conditions for when the policy is in effect.

Protection Policy: Enforced through MFA (Multi-Factor Authentication) and a strong Password Policy.

MFA Options

- Google Authenticator (virtual MFA)
- Authy (virtual MFA)
- YubiKey (3rd-party hardware security key)
- Hardware Key Fob Device
- SurePass ID — AWS GovCloud

How Users Can Access AWS

Access Method	Protected By
AWS Management Console	Password + MFA
AWS Command Line Interface (CLI)	Access Keys
AWS Software Development Kit (SDK)	Access Keys

- Access Keys are generated through the console; each user manages their own keys.
- **Never share access keys.**

AWS CLI (Command Line Interface)

- Lets you interact with AWS services using commands in a command-line shell.
- Gives direct access to the public APIs of AWS services.
- Used to manage resources via scripts — open source, and an alternative to the Management Console.

AWS SDK (Software Development Kit)

- Language-specific APIs — a set of libraries, with a dedicated SDK for each supported programming language.
- Another alternative to the Management Console: manage and access AWS services through code.
- Not used within a terminal — it must be embedded within your application.
- Supports a wide range of programming languages.
- The AWS CLI itself is built on top of the AWS SDK for Python.

AWS CloudShell

- Not available in every region.
- A terminal built into the AWS Console — functionally similar to the CLI.
- Essentially Linux with AWS-specific commands pre-configured.

PART 4 · IAM ROLES, AUDITING & BEST PRACTICES

IAM Roles for AWS Services

- AWS services often need to perform actions/tasks on your behalf.
- Permissions are granted to those services using **IAM Roles**.
- An EC2 Instance is a virtual server — and a good example of this pattern.

Example flow: An EC2 instance needs to perform an action in AWS. The instance is given an IAM Role. When the instance makes an API call, AWS checks the role's permissions — if they're correct, AWS responds to the call.

IAM Security Tools

Tool	Level	What It Does
IAM Credentials Report	Account-level	Lists all users in the account and the status of their credentials.
IAM Access Advisor	User-level	Shows the service permissions granted to a user and when each was last accessed — useful for trimming unused permissions to align with least privilege.

IAM Guidelines & Best Practices

- Don't use the root user account except for initial account setup.
- 1 physical user = 1 AWS user (don't share accounts).
- Assign users to groups, and assign permissions at the group level.
- Enforce a strong password policy.
- Use and enforce MFA.
- Create and use IAM Roles to grant permissions to AWS services.
- Use Access Keys only for programmatic access (CLI/SDK), and keep them protected.
- Use the IAM Credentials Report and Access Advisor's last-accessed data to regularly audit account permissions.
- **Never share IAM users or access keys.**